

Pythagoras' Theorem & Trigonometry

Extra practice test 3 — answer sheet · 27 marks total

Section A — Pythagoras' theorem

A1. $a^2 + b^2 = 8^2 + 15^2 = 64 + 225 = 289$ (1) $c = \sqrt{289} = \mathbf{17\text{ cm}}$ (1)

A2. $a^2 = 25^2 - 7^2$ (1) $= 625 - 49 = 576$ (1) $a = \sqrt{576} = \mathbf{24\text{ cm}}$ (1)

A3. $\text{diagonal}^2 = 20^2 + 21^2 = 400 + 441 = 841$ (1) $\text{diagonal} = \sqrt{841} = \mathbf{29\text{ cm}}$ (1)

Section B — SOH CAH TOA basics

B1. **TOA** (1)

B2. **SOH**: $\text{hypotenuse} = \text{opposite} \div \sin \theta$ (1)

B3. **CAH**: $\theta = \cos^{-1}(\text{adjacent} \div \text{hypotenuse})$ (1)

Section C — Finding missing sides

C1. $\sin 48^\circ = x \div 14$ (1) $x = 14 \times \sin 48^\circ$ (1) $= \mathbf{10.4\text{ cm}}$ (1)

C2. $\cos 29^\circ = 11 \div h \rightarrow h = 11 \div \cos 29^\circ$ (1) $= \mathbf{12.6\text{ cm}}$ (1)

C3. $\tan 51^\circ = x \div 4 \rightarrow x = 4 \times \tan 51^\circ$ (1) $= \mathbf{4.9\text{ cm}}$ (1)

Section D — Finding missing angles

D1. $\sin \theta = 5 \div 12 = 0.4167 \rightarrow \theta = \sin^{-1}(0.4167)$ (1) $= \mathbf{24.6^\circ}$ (1)

D2. $\tan \theta = 6 \div 8 = 0.75$ (1) $\theta = \tan^{-1}(0.75)$ (1) $= \mathbf{36.9^\circ}$ (1)

Section E — Applying Pythagoras & trigonometry

E1. Use Pythagoras: $\text{height}^2 = 4^2 - 1.5^2 = 16 - 2.25 = 13.75$ (1) $\sqrt{13.75} = \mathbf{3.7\text{ m}}$ (1)

E2. $\sin 40^\circ = h \div 25$ (1) $h = 25 \times \sin 40^\circ$ (1) $= \mathbf{16.1\text{ m}}$ (1)

Marking note: award method marks for a correctly set-up equation or calculation even if the final answer contains a rounding or arithmetic slip. Accept any correct rounding to 1 decimal place unless a different level of accuracy is specified. Missing units on an otherwise-correct numerical answer should lose the final accuracy mark, not the method mark.